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5. Constructor in Java

5.1 Constructor :

In Java, a constructor is a block of codes similar to the method. It is called when an instance of the class is created. At the time of calling constructor, memory for the object is allocated in the memory.

It is a special type of method which is used to initialize the object. Every time an object is created using the new() keyword, at least one constructor is called. It calls a default constructor if there is no constructor available in the class.

Rules For creating Java constructor :

1. Constructor name must be the same as its class name.
2. A Constructor must have no explicit return type
3. A Java constructor cannot be abstract, static, final and synchronized.

5.2 Constructor Types :

There are two types of constructor in Java.

1. Default constructor (no-arg)
2. Parameterized constructor

Java default constructor :

A constructor is called "Default constructor" when it doesn't have any parameter.

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Syntax:

```
<class_name>(){}
```

Example:

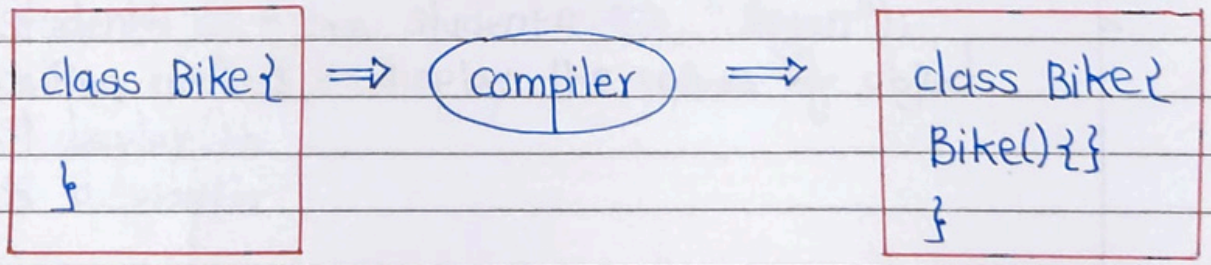
```
// Java program to create and call a default constructor
class Bike1{
// creating a default constructor
Bike1(){ System.out.println("Bike is created");}
// main method
public static void main(String args[]){
// calling a default constructor.
Bike1 b = new Bike1();
}
}
```

output:

Bike is created.

Rule:

If there is no constructor in a class, compiler automatically creates a default constructor.



The default constructor is used to provide the default values to the object like 0, null, etc..., depending on the type.

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Java parameterized constructor:

A constructor which has a specific number of parameters is called parameterized constructor. The parameterized constructor is used to provide different values to distinct objects. However, you can provide the same values too.

Example:

```
// Java program to demonstrate the use of the parameterized constructor.
class Student4 {
  int id;
  String name;
 // creating a parameterized constructor
  Student4(int i, String n) {
  id = i;
  name = n;
  }
 // method to display the values
  void display() { System.out.println(id+" "+name);}
  public static void main (String args []) {
 // creating objects and passing values
   Student4 s1 = new Student4 (111, "karan");
   Student4 s2 = new Student4 (222, "Aryan");
 // calling method to display the values of object
   s1.display();
   s2.display();
  }
 }
```

Output:

```
111 karan
222 Aryan
```

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5.3 Constructor Overloading in Java

In Java, a constructor is just like a method but without return type. It can also be overloaded like Java methods.

Constructor Overloading in Java is a technique of having more than one constructor with different parameter lists. They are arranged in a way that each constructor performs a different task. They are differentiated by the compiler by the number of parameters in the list and their types.

Example of constructor overloading :

```
public class Student {
    string name;
    int age;
    // constructor with one parameter
    public Student (string name){
        this.name = name;
    }
    // Constructor with two parameters
    public Student (string name, int age){
        this.name = name;
        this.age = age;
    }
}
```

5.4 static keyword :

The static keyword in java is used for the memory management mainly. We can apply static keyword.

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with variables, methods, blocks and nested classes. The static keyword belongs to the class than an instance of the class.

The static can be:

1. Variable (also known as a class variable)
2. Method (also known as a class method)
3. Block
4. Nested class.

Java static variable:

If you dedare any variable as static, it is known as a static variable. The static variable can be used to refer the common property of all objects (which is not unique for each object) for ex: the company name of employes, college name of students, etc.

The static variable gets memory only once in the class area at the time of class loading

Java static Method:

- A static method belongs to the class rather than object of a class
- A static method can be invoked without the need for creating an instance of a class.
- A static method can access static data member and can change the value of it.

Java static block:

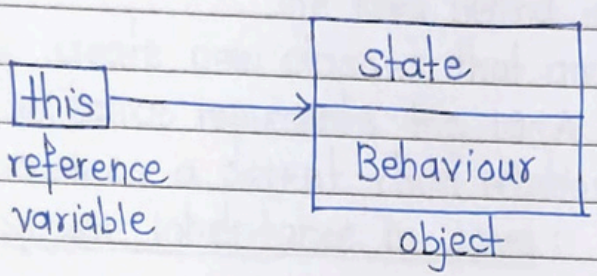
- Is used to initialize the static data member.

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- It is executed before the main method at the time of class boarding.

5.5 this keyword in Java :

There can be a lot of usage of Java this keyword. In Java, this is a reference variable that refers to the curren object.



- this can be used to refer current class instance variable
- this can be used to invoke current class method.
- this() can be used to invoke current class constructor.
- this can be passed as an argument in the method call.
- this can be passed as an argument in the constructor call.
- this can be used as return the current class instance from the method

Example:

```
Public class Person{
  string name;
  public Person (string name){
    this.name = name; // "this.name" refers to instance variable
  }
  Public void printname(){
    System.out.println("Name:" + this.name);}}
```